Supplementary Material

Ping Yang[1], Tianmiao Chen[1,*], ZhuoLin Zhong[1], Yuhang Wu[1], and Qingquan Luo[1]

[1]School of Electric Power Engineering, South China University of Technology, Guangzhou, 510000, China
[*]202410183266@mail.scut.edu.cn

ABSTRACT

Data centers require intraday scheduling that balances energy costs with renewable consumption. However, existing methods struggle with multi-source uncertainties in forecasting electricity prices, rooftop PV generation, and task arrivals. We propose a decision-focused online learning optimal scheduling method, which embeds the downstream optimization into forecasting model training, using decision loss rather than prediction error as the objective so forecasts directly serve scheduling quality. The method integrates forecasts of electricity prices, PV generation, and task arrivals: day-ahead training on historical samples captures long-term trends, while intraday rolling execution with delayed parameter updates — executing only the first action per step and updating after true values are observed — fully exploits real-time information. A regularized surrogate loss handles the non-differentiability of storage charging–discharging complementarity constraints during backpropagation. Experiments on real operational data from a data center cluster (June–December 2025) show monthly energy cost reductions of 16.5%–32.4% over conventional predict-then-decide methods, and cumulative PV consumption improving from 64.1% to 78.3% over seven months, confirming that decision-focused online learning effectively leverages real-time measurements to continuously improve both scheduling economy and renewable energy utilization.

Supplementary Material: Input Feature Description

This supplementary material gives the input feature construction of the three forecasting models in Section 3.4 of the main text. All continuous features are standardized with training-set statistics, and the validation set, test set, and intraday online stage use the same mean and standard deviation to avoid future information leakage. For the q-th continuous feature of any module a,

$$\tilde{x}_{t,u}^{a,q} = \frac{x_{t,u}^{a,q} - \mu_{\mathrm{tr}}^{a,q}}{\sigma_{\mathrm{tr}}^{a,q}}. \tag{1}$$

The computing task forecasting module takes the historical task arrivals of the node and calendar variables as inputs. Let $\lambda_{t,u}$ denote the task arrival volume observed at node u in period t, with the lag set

$$\mathscr{L}_{dc} = \{1,2,3,4,8,16,96,192,672\}, \tag{2}$$

where the lag step is measured in 15 min sampling points. Let $m_t \in [0,1440)$ be the intraday minute and $q_t \in \{0,1,\dots,6\}$ the weekday index; then the calendar features are

$$\begin{aligned} c_t^{dc} = \Big[&\sin\left(\frac{2\pi m_t}{1440}\right), \cos\left(\frac{2\pi m_t}{1440}\right), \\ &\sin\left(\frac{2\pi q_t}{7}\right), \cos\left(\frac{2\pi q_t}{7}\right), \mathbf{1}\{q_t \geq 5\}\Big]^\top \in \mathbf{R}^5. \end{aligned} \tag{3}$$

The computing task input vector is

$$x_{t,u}^{dc} = \left[\{\lambda_{t-\ell,u}\}_{\ell \in \mathscr{L}_{dc}}, (c_t^{dc})^\top\right]^\top \in \mathbf{R}^{14}, \tag{4}$$

consisting of 9 lag features and 5 calendar features. The model output is restored by $\mathscr{T}_{dc}(\cdot)$ and clipped to non-negative task arrival volumes.

The available renewable power forecasting module provides the upper bound $P_{t,u}^R$ of renewable power that can be dispatched and consumed at node u in period t. The few negative values in the raw PV output P_t^R are first clipped to $P_t^{R,+} = \max\{P_t^R, 0\}$, and the training labels are normalized by the training-set baseline capacity:

$$y_t^R = \frac{P_t^{R,+}}{P_{\mathrm{base}}^R}, \qquad P_{\mathrm{base}}^R = \max_{t \in \mathscr{S}_{\mathrm{tr}}} P_t^{R,+}. \tag{5}$$

Considering the unified linear forecasting model, the NWP meteorological inputs take the two channels (radiation flux and temperature) at the grid points of the PV stations of each node. If the raw NWP array is $Z^R_{t,i,j,k}$, then

$$n^R_{t,u} = \left[Z^R_{t,i_u,j_u,1}, Z^R_{t,i_u,j_u,2}\right]^\top \in \mathbf{R}^2, \tag{6}$$

where $(i_u, j_u) \in \{(2,2),(10,10)\}$ is the position of the PV station of node u in the NWP grid. The renewable calendar and solar position features are

$$\begin{aligned} c^R_t = \big[&\sin\left(\frac{2\pi m_t}{1440}\right), \cos\left(\frac{2\pi m_t}{1440}\right), \sin\left(\frac{2\pi r_t}{12}\right), \cos\left(\frac{2\pi r_t}{12}\right), \\ &\sin h_t, \cos h_t, \sin\alpha_t, \cos\alpha_t\big]^\top \in \mathbf{R}^8, \end{aligned} \tag{7}$$

where r_t is the month, h_t the solar elevation angle, and α_t the solar azimuth angle. The clear-sky theoretical power is normalized as $\tilde{P}^{R,\mathrm{cs}}_t = P^{R,\mathrm{cs}}_t / P^R_{\mathrm{base}}$ and used as an exogenous power covariate. The renewable input vector is

$$x^R_{t,u} = \left[(n^R_{t,u})^\top, (c^R_t)^\top, \tilde{P}^{R,\mathrm{cs}}_t\right]^\top \in \mathbf{R}^{11}. \tag{8}$$

In the intraday rolling window, $X^R_{t:t+H-1,u} = [(x^R_{t,u})^\top, \dots, (x^R_{t+H-1,u})^\top]^\top$ is constructed from the current time onward and restored to the physical power range by $\mathscr{T}_R(\cdot)$.

The electricity price forecasting module generates the market purchase and sale prices within the scheduling window. Let $\pi_{t,u}$ denote the reference electricity price of region u in period t, and d_t the total regional demand in the same period. The price calendar features add a monthly periodicity term on the basis of Equation (3):

$$\begin{aligned} c^\pi_t = \big[&\sin\left(\frac{2\pi m_t}{1440}\right), \cos\left(\frac{2\pi m_t}{1440}\right), \sin\left(\frac{2\pi q_t}{7}\right), \cos\left(\frac{2\pi q_t}{7}\right), \\ &\sin\left(\frac{2\pi r_t}{12}\right), \cos\left(\frac{2\pi r_t}{12}\right), \mathbf{1}\{q_t \geq 5\}\big]^\top \in \mathbf{R}^7. \end{aligned} \tag{9}$$

The price lag set and demand lag set are

$$\begin{aligned} &\mathscr{L}_\pi = \{0,1,2,3,4,8,16,96,288\}, \\ &\mathscr{L}_d = \{0,1,2,4,8,16,96,288\}. \end{aligned} \tag{10}$$

where lag 96 denotes the same time on the previous day and lag 288 the same time three days earlier (in units of 15 min periods). To characterize short-term fluctuations and daily-periodic levels, the rolling statistics window $\mathscr{R} = \{4, 96\}$ is set, and the price rolling mean $\bar{\pi}_{t,r}$, price rolling standard deviation $\sigma_{\pi,t,r}$, and demand rolling mean $\bar{d}_{t,r}$ are constructed. The price forecasting input vector is

$$\begin{aligned} x^\pi_{t,u} = \big[&(c^\pi_t)^\top, \{\pi_{t-\ell,u}\}_{\ell\in\mathscr{L}_\pi}, \{d_{t-\ell}\}_{\ell\in\mathscr{L}_d}, \\ &\{\bar{\pi}_{t,r}, \sigma_{\pi,t,r}, \bar{d}_{t,r}\}_{r\in\mathscr{R}}\big]^\top \in \mathbf{R}^{30}. \end{aligned} \tag{11}$$

The above lag and rolling statistics use only historical observations at and before t; the day-ahead forecasting uses only information available from the previous day and earlier. Since spot electricity prices can be negative and may spike, $\mathscr{T}_\pi(\cdot)$ performs only scale restoration without non-negative clipping.